CHAPTER 6

Non-Infectious Pathology in Asian Elephants

Poisoning, Electrocution, Reproductive & Metabolic Disorders

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1. Elephant Poisoning in India — Overview

Poisoning is one of the most severe threats to **pachyderms** — the large, thick-skinned mammals including elephants, rhinoceroses, and hippopotamuses. In India, unnatural causes such as electrocution and poisoning claim hundreds of elephant lives every year. Incidents arise primarily from human–wildlife conflict and illegal poaching, and parallel patterns are seen in Africa.

Route	Examples & Context
Intentional	Illegal poaching; retaliatory killing in human–elephant conflict areas. Deliberate placement of poisoned bait or contaminated food near crop fields.
Accidental	Elephants consuming mycotoxin-infected crops (e.g. kodo millet); drinking from algae-bloom water bodies; ingesting toxic plants unknowingly.

2. Mycotoxins & Biological Toxins

A. Crop-Borne Mycotoxins (CPA / Kodo Millet)

Elephants raiding farm fields are highly susceptible to **mycotoxins** — secondary metabolites of moulds. The most clinically significant is **Cyclopiazonic Acid (CPA)**, produced by *Aspergillus* and *Penicillium* species infecting **kodo millet** crops.

Pathological effects: acute vascular damage, liver necrosis, and kidney necrosis, ultimately leading to multi-organ failure and death.

> **Bandhavgarh Case Reference**
> Key case — Bandhavgarh Tiger Reserve, MP: A herd of 13 elephants raided kodo millet crops; 10 elephants died. Forensic post-mortem and lab investigation confirmed accidental toxicity from fungal mycotoxins in infected crop.
> (Speaker note: 2 similar cases personally seen/treated in KTR camp elephants, 1997–98.)

When Does Contamination Occur?

Heavy or unseasonal rainfall coinciding with grain maturation/harvest creates moist, warm conditions ideal for fungal colonisation. Unharvested grain in standing fields carries the highest risk.

Management & Prevention:

1. **Good agricultural practices** — correct spacing, timely harvesting, and minimising field duration after maturity.
2. **Biocontrol agents** — application of non-toxigenic fungal strains (*Aspergillus* spp.) to competitively exclude toxin-producing strains.

3. **Post-harvest handling** — thorough drying and airtight storage to prevent moisture-driven fungal re-growth.

B. Cyanotoxins / Cyanobacteria

Toxic blue-green algae (cyanobacteria) proliferate in stagnant, warm, nutrient-rich water bodies during **hot, dry seasons**. Dense blooms produce potent **neurotoxins and hepatotoxins** that can cause fatal mass die-offs — especially at elephant watering holes where large volumes of contaminated water are consumed rapidly.

C. Toxic Plants

Elephants generally avoid toxic vegetation through learned behaviour, but unfamiliar environments, food scarcity, or captivity can lead to accidental ingestion. Key examples:

Plant / Group	Toxic Mechanism & Effect	System Affected
Oleander (*Nerium oleander*)	Cardiac glycosides → fatal arrhythmia and cardiac arrest	Cardiac
Certain weeds / unfamiliar plants	Variable alkaloids, glycosides → neurological signs	Neurological
Calcium oxalate plants (e.g. *Dieffenbachia*, aroids)	Crystal penetration of oral/GI mucosa → intense swelling, irritation, and systemic distress if eaten in quantity	Mechanical + systemic

3. Hydrocyanic Acid (HCN) Poisoning

Source & Risk Factors

Cyanogenic glycosides in **jowar / sorghum (*Sorghum bicolor*)** are hydrolysed to release HCN (prussic acid). Risk is highest in:

- **Young, rapidly growing plants** — glycoside content peaks before maturity.
- **Stunted or stressed plants** — e.g. following drought then autumn rains.
- **Leaves and stalks** — contain far more glycoside than grain.
- Feeding elephants **large, unfamiliar quantities** of fresh jowar at once (especially when regular feed is unavailable).

Case Study — 5 Captive Elephants (Circus, Jabalpur)

Reference: Zoo Print, Vol. XXV, 6 June 2010.

Five circus elephants were fed large quantities of fresh jowar (unavailability of regular feed). They developed **restlessness and diarrhoea**; routine anti-diarrhoeal treatment failed and condition worsened. HCN poisoning was suspected and the following treatment was instituted:

Step	Action
Step 1	Stopped jowar feeding immediately.
Step 2	Dextrose saline (8–10 L) + 30 mL multivitamin injection (MVI) via ear vein.
Step 3	Sodium thiosulphate (50 g) dissolved in water, given orally as antidote.
Step 4	Treatment repeated at approximately 12-hour intervals.

Step	Action
Step 5 (critical cases)	Additional doses + cold dextrose saline + small ice pieces + Tab. Diadin per rectum.
Outcome	Recovery achieved; all animals resumed drinking and grazing.

Key Learning Point

Lesson: HCN poisoning here was accidental — caused by feeding large amounts of unfamiliar sorghum to elephants not accustomed to it, due to non-availability of regular food. Sodium thiosulphate is the key antidote (converts cyanide to thiocyanate).

4. Electrocution

Electrocution is death or severe injury caused by an electric shock. The body becomes part of an electrical circuit; high voltage or current disrupts the body's natural electrical systems, commonly triggering **cardiac arrest or severe electrical burns**.

Term	Definition
Electric Shock	Physical sensation or injury from electricity — muscle spasms, minor burns. NON-FATAL.
Electrocution	An electric shock that is FATAL. The term implies death as the outcome.

A documented case of elephant electrocution was recorded in November 2020.

5. Pyometra

Definition & Affected Population

Pyometra is a severe, life-threatening **uterine infection** marked by accumulation of pus and suppurative inflammation of the uterus. It occurs most commonly in **aging, nulliparous (never-given-birth) female elephants** and is linked to prolonged hormonal fluctuations and underlying reproductive issues.

Causes & Pathophysiology

Factor	Mechanism / Details
Hormonal changes	Fluctuating progesterone levels → abnormally thickened uterine lining (endometrial hyperplasia) creates a nutrient-rich environment for bacterial growth.
Bacterial infection	Secondary invasion by organisms such as *E. coli* or *Enterococcus faecium* → suppurative inflammation and pus accumulation.
Pre-existing conditions	Cystic endometrial hyperplasia or benign tumours (leiomyomas) in older captive females predispose to pyometra.

Treatment

1. **Hormonal downregulation** — GnRH vaccines (e.g. Improvac) or Deslorelin implants to suppress reproductive cycles and reduce endometrial stimulation.
2. **Aggressive antibiotic therapy** — broad-spectrum systemic antibiotics targeting specific isolated pathogens.

3. **Uterine lavage** — flushing and draining the uterus to remove accumulated pus.
4. **Homeopathic adjunct protocol practised:** Pyrogenium 1000 + Hepar Sulph 1000 + Secale 1000.

Surgical management: vaginal vestibulotomy has been performed in Asian elephant cases.

6. Hyperthermia

Why Elephants Are Prone to Overheating

Hyperthermia is a life-threatening condition arising from high environmental temperature, lack of access to water/shade, massive body volume, and the absence of conventional sweat glands. Three converging factors explain the heightened risk in elephants:

Risk Factor	Explanation
Square-Cube Law	As body volume increases, metabolic heat production scales faster than the relatively smaller skin surface area available to dissipate it. Large size = excellent heat retention, poor heat loss.
No sweat glands	Elephants lack traditional eccrine sweat glands; only pores between the toes are present. They depend on transepidermal water loss through the skin — which requires ready access to drinking water.
Continuous metabolic heat	Sustained walking or exertion in direct sun generates metabolic heat faster than passive cooling can remove it, triggering hyperthermia in the absence of active cooling behaviour.

Natural Cooling Mechanisms

- **Ear flapping** — the large, heavily vascularised pinnae act as thermal radiators. Rapid flapping increases convective heat loss from blood circulating through the ears, lowering overall body temperature.
- **Wallowing and water/mud spraying** — immersion or spraying with water provides powerful evaporative cooling; a residual mud coat shields the skin from solar radiation (natural sunscreen).
- **Saliva spraying** — self-spraying with trunk-collected saliva adds evaporative surface cooling.
- **Increased skin permeability** — the hide becomes more permeable during heat stress, allowing greater transepidermal evaporative cooling — but only if adequate drinking water is available.

Management of Hyperthermia

Intervention: Provide unlimited cool drinking water and shade immediately.
Apply cold water sprays to the skin, behind the ears, and on the feet.
Move the animal out of direct sun; reduce physical exertion.
In severe cases: cold dextrose saline IV, ice packs to ear margins.

7. Iron Deficiency Anaemia

Clinical Features & Diagnosis

Definition: Qualitative and quantitative reduction in circulating red blood cells and haemoglobin. In elephants, chronic dietary deficiency or parasitism are common underlying causes.

Finding	Significance
Pale mucous membranes	Reduced haemoglobin → less colour in conjunctiva, oral mucosa

Finding	Significance
Extreme weakness	Reduced oxygen-carrying capacity → poor cellular energy production
Inability to walk / slow movement	Muscle hypoxia; severe cases cannot rise
Blood smear — target cells	RBCs show central pallor with dense peripheral rim ("target cell" morphology
Microcytic hypochromic RBCs	Small, pale cells on haematology — hallmark of iron-deficiency anaemia
Low haemoglobin (Hb)	Confirmed on complete blood count (CBC)

Treatment

1. **Diet restoration** — full normal elephant ration: soyabeans, horse gram, napier grass, bamboo, leaves, grains, seasonal fruits.
2. **Iron supplementation** — natural sources: jaggery (rich in iron), honey; plus oral or parenteral iron preparations as indicated.
3. **Vitamin B12 injections** — essential cofactor for RBC maturation.
4. **Multivitamin and mineral mixture** supplementation to correct concurrent micro-nutrient gaps.

8. Quick Reference — Conditions Covered

Condition	Category	Key Agent / Trigger	Hallmark Finding
Kodo millet poisoning	Toxic / mycotoxin	CPA from *Aspergillus*/*Penicillium*	Liver/kidney necrosis, cardiac failure, rapid death
HCN poisoning	Toxic / plant	Cyanogenic glycosides in sorghum/jowar	Restlessness, diarrhoea, respiratory distress
Cyanotoxins	Toxic / environmental	Cyanobacterial blooms in stagnant water	Neuro- and hepatotoxicosis; mass die-offs
Plant toxicity	Toxic / plant	Oleander (cardiac), calcium oxalate plants	Cardiac arrhythmia, neurological signs, GI irritation
Electrocution	Physical	Power-line / electrified-fence contact	Cardiac arrest, burns; rapid decomposition post-mortem
Pyometra	Reproductive	Hormonal + bacterial (uterine)	Uterine pus, systemic sepsis; older captive females
Hyperthermia	Thermal / metabolic	High temperature + no sweat glands	Collapse, organ failure; respond to cooling measures
Iron deficiency anaemia	Nutritional	Poor diet, parasitism, chronic disease	Pale mucosae, weakness, microcytic hypochromic RBCs

Key Takeaways

Chapter Summary — Non-Infectious Pathology in Asian Elephants

1. Elephant poisoning is both intentional (retaliation/poaching) and accidental (mycotoxins, toxic plants, algal blooms) — field history is critical to distinguish.
2. CPA from kodo millet is the most documented mycotoxin killing wild elephants in India; heavy rainfall + unharvested grain = highest risk.
3. HCN from jowar is treatable with sodium thiosulphate — suspect when standard anti-diarrhoeal therapy fails after recent sorghum feeding.
4. Electrocution means death; distinguish from non-fatal electric shock at necropsy (burn marks, rapid decomposition, absent rigor mortis).
5. Pyometra targets aging, nulliparous captive females — hormonal + antibiotic treatment; surgical vestibulotomy in severe cases.
6. Hyperthermia is unique to large-bodied, sweat-gland-deficient animals; first response is water, shade, and active skin cooling.
7. Iron deficiency anaemia: confirm with blood smear (target cells, microcytic hypochromic RBCs); treat with diet correction + iron + B12.